ABSTRACT OF THE DISCLOSURE

A thermal management system for a solid-state battery pack places phase-change modules between adjacent cells and couples neighbouring modules with bidirectional thermoelectric bridges. A controller predicts each cell's temperature trajectory with a reduced-order model and drives the bridges so that heat moves from cells predicted to overheat to cells predicted to fall below their optimal window. Heat stored during fast charging is returned to the cells during cold starts, narrowing cell-to-cell temperature spread and reducing capacity fade.